Counting fills misrepresents liquidations: evidence from a complete on-chain derivatives ledger

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Abstract

Forced liquidations in cryptocurrency derivatives are almost always counted from exchange-published event feeds, in which one economic liquidation appears as several records. We quantify the resulting distortion on Hyperliquid, an on-chain perpetual-futures venue whose node-fills archive is, to our knowledge, the only public liquidation record that is simultaneously complete and attributed to the liquidated account. On a uniformly stratified sample of 351,540 liquidation episodes reconstructed from 2,010,042 fills over one year, counting fills rather than episodes inflates the event count by a factor of **5.72**. The inflation is not a constant: it is *size-dependent*, rising from a median of 2 fills per episode below the median size to a median of 72 in the top percentile, so the bias is concentrated on precisely the events that liquidation studies are about. The top 1% of episodes generate 23.1% of all fills. We further characterise the episode-size distribution as far as the data permits, and report a negative result that we believe is the more useful one: the tail is unambiguously heavy—an exponential is rejected decisively—and it is not a power law, but it *cannot be named* beyond that. Lognormal and Weibull specifications both dominate a Pareto fit at every estimable threshold across nearly three orders of magnitude of cut-off, yet which of the two wins reverses with the threshold, so any named family would report the analyst's cut-off rather than the distribution. Finally, we place this beside the liquidation-data bias that is already documented and observe that it runs the *other way*: rate-limited centralised-exchange feeds undercount, publishing per-second maxima rather than samples, while fill-level on-chain records overcount. Pooling the two sources combines a downward-biased count with an upward-biased one, and they do not cancel. All data, code and pre-registrations are public.

1 Introduction

Empirical work on cryptocurrency derivatives routinely counts liquidation events: how many occurred, how large they were, how they cluster in cascades, how they propagate across venues. The counting is nearly always done from an exchange-published liquidation feed, and the unit of that feed is not the unit of the question.

A trader whose position is force-closed experiences one liquidation. The exchange executes it as a sequence of fills against the order book. If the published record is per fill, then a single economic event enters the dataset several times, at several sizes, none of them the size of the position that was closed. Any statistic computed over those records—a count, a size distribution, a tail index, an inter-arrival time—measures the exchange's execution mechanics as much as the market.

This is a well-understood problem in principle. What has been missing is a public dataset in which the correct unit can be reconstructed, so that the distortion can be measured rather than assumed. That requires a liquidation record that is (i) complete, with no sampling or rate limit, and (ii) attributed, so that fills belonging to one account's forced close can be grouped. To our knowledge no centralised venue publishes both.

Hyperliquid does. It is an on-chain perpetual-futures exchange whose node data includes every fill, each carrying the liquidated account address, the mark price at liquidation, the starting position and the realised loss. This paper uses that archive to measure how far fill-counting departs from episode-counting, and what the episode-size distribution actually looks like.

Contributions.

1. We measure a **5.72**× count inflation from fill-counting on a uniformly stratified one-year sample, and show the factor is robust to how "one liquidation" is defined (Section 4).
2. We show the inflation is **size-dependent**, growing from 2 to 72 fills per episode between the lower half and the top percentile of the size distribution, so it is worst exactly where liquidation research concentrates (Section 5).
3. We show fill-counting **compresses the size distribution**, by factors of 4.58× at the 99th percentile and 10.0× at the 99.9th, and that the two instrument segments compress differently (Section 6).
4. We characterise the episode-size tail and report that it **cannot be named**: heavy, not a power law, but with the ranking between lognormal and Weibull reversing across the threshold range (Section 7).
5. We document what centralised-exchange feeds actually publish, and show the failure mode there is different and worse for counting purposes (Section 8).
6. We position this against the liquidation-data bias already documented in the literature, which runs in the *opposite* direction, and note the consequence for work that pools centralised and on-chain liquidation records (Section 2).

On the negative result. Section 7 sets out to name the tail and fails. We report the failure in full because the alternative—quoting whichever family wins at the threshold one happened to choose—is how a threshold choice becomes a stylised fact. Three tail-index estimates were produced during this work and all three were withdrawn; the record of those withdrawals is public and is described in Section 11.

2 Related work

2.1 The known bias runs the other way

That published liquidation counts are unreliable is not a new observation. What has been documented is the opposite of what we measure.

Since 2021 the major centralised venues have rate-limited their public liquidation feeds. Binance publishes only the largest liquidation order per 1000 ms per symbol; OKX publishes at most one update per second per contract; Bybit's legacy stream did the same before `allLiquidation` replaced it in 2025 [13, 14, 15]. K33 Research [11] characterised the consequence bluntly—their analyst described exchange liquidation data as "a vast underrepresentation of actual liquidation volumes"—and industry commentary during the October 2025 cascade reported venue-internal figures several times the published ones.

The direction matters. Rate-limited order feeds **undercount**. Fill-level records, which is what an on-chain venue publishes, **overcount**. Both defects are real, they apply to different data sources, and they pull in opposite directions. A study that pools centralised-feed liquidations with on-chain liquidations is combining a downward-biased count with an upward-biased one, and the two do not cancel.

A third source in common use, aggregator heatmaps, is neither: those are *modelled* estimates derived from open-interest snapshots and assumed leverage distributions, not records of liquidations that occurred [16].

2.2 What the empirical literature counts

Perpetual futures. Cheng et al. [4] give the reference empirical treatment of forced liquidation in crypto perpetuals, applying generalised extreme value theory to BitMEX data from 2017–2021 and reporting daily forced liquidations of 3.51% and 1.89% of outstanding long and short positions, at an average leverage of 60× among liquidated accounts. Their statistics are *volumes* rather than event counts, which insulates them from the unit question we raise—aggregate liquidated notional is invariant to how the record is partitioned—though a size distribution drawn from the same feed would not be.

Recent work on the October 2025 cascade is event-study shaped and likewise measures in dollars: Lim [9] works minute-level on Binance and Bybit, and Chitra et al. [7] analyse Hyperliquid's autodeleveraging during the same episode, quantifying profit overshoot (up to $51.7M) rather than counting discrete events. Sepper [8] builds a liquidity-risk framework on Hyperliquid order-book and liquidation records. Zhivkov et al. [10] compare microstructure across 26 centralised and decentralised venues.

To our knowledge, none of this work counts liquidation *events* from a fill-level record, and none states the unit at which liquidations are counted. We therefore do not claim that any published result is wrong. We claim that the correction has not been measured, and we measure it.

Lending protocols. The DeFi lending literature is the closest methodological neighbour and instructive precisely because it does not have this problem. Qin et al. [5] study roughly 50,000 liquidations across Aave, Compound, MakerDAO and dYdX by scanning for protocol-emitted events—Aave's `LiquidationCall`, Compound's `LiquidateBorrow`—and Moallemi and Patange [6] analyse fixed-spread liquidation on the same event basis. In lending, **the protocol emits one event per liquidation call, so the unit is given by the data**.

Perpetual venues emit no such event. A forced close is observable only through the fills it generates, so the unit must be *reconstructed* rather than read. That is the gap this paper addresses: the reconstruction is a researcher degree of freedom that has not been priced.

3 Setting and data

3.1 Hyperliquid liquidations

Hyperliquid is a perpetual-futures venue with an on-chain order book. Positions are margined against a mark price, and a position whose margin ratio falls below maintenance—half the initial margin at maximum leverage, between 1.25% and 16.7% depending on the instrument—is force-closed by the protocol. The resulting trades are recorded in the node's fill stream with a `liquidation` field carrying the liquidated user, the mark price and the liquidation method, alongside the fill's own price, size, starting position and realised profit and loss.

Two mechanism details bear directly on the measurement. First, **large positions are closed in parts**: above 100,000 USDC of position value the protocol sends only 20% of the position as a market order, then waits **30 seconds** before sending more for that account [12]. Second, if the position is still not closed and equity falls below two-thirds of maintenance margin, a *backstop* transfers the position to the liquidator vault.

The first of these is a documented mechanism for the size dependence we report in Section 5: tranching is not only order-book slicing but partly protocol design. It also raises a measurement question—whether chunks separated by the cooldown should be one liquidation or several—which we test directly in Section 5.2.

Two properties make the archive suitable for this measurement. It is **complete**: every fill is present, with no publication rate limit, because the data is the chain's own record rather than a broadcast feed. And it is **attributed**: each fill names the liquidated account, so fills belonging to one forced close can be grouped without inference.

The archive is hosted as requester-pays object storage, partitioned by hour.

3.2 Sampling design

A full census of the archive year is approximately 8,760 hours and on the order of 300 GB of egress. We instead sample **four fixed hours per day—02, 08, 14 and 20 UTC—across all 365 archive days**, which is uniform in time and inside the free egress allowance.

The hours are *fixed rather than chosen*. This matters more than the sample size. An earlier version of this measurement used twelve hours, two of which had been selected because they contained known liquidation cascades, and therefore over-represented large episodes in a study whose subject is the size dependence of a bias. Fixing the hours removes that selection directly rather than correcting for it afterwards.

The sample comprises **351,540 episodes** reconstructed from **2,010,042 fills**, across **380 instruments** and **151,730 distinct accounts**, totalling **$15.53 billion** of liquidated notional.

Two passes. The counting results below come from that collection, which reduced to episodes. Measuring compression needs the individual fill notionals, which it did not retain, so the same hours were collected a second time keeping them. The second pass returned 351,648 episodes and 2,010,314 fills—**0.03% more**, most likely late-arriving blocks written after the first pass read those hours—and reproduces every count statistic reported in Sections 5 to three decimal places: the inflation factor, the median and mean fills per episode in the top percentile, the share of fills and of notional carried by that percentile, the rank correlation between size and fills, and the largest episode and its fill count. We therefore quote the first collection for counts and the second for compression, and note the difference rather than reconciling it away.

3.3 Instrument segments

Hyperliquid hosts both venue-listed perpetuals ("majors") and builder-deployed markets under its HIP-3 mechanism, identified by a `dex:` prefix in the instrument name. HIP-3 markets are thinner and separately parameterised, so we report them separately throughout: they carry 17.7% of episodes but 9.3% of notional—more liquidations, smaller.

4 What counts as one liquidation

The natural unit is the *episode*: one account's forced close of one position. The archive does not label episodes, so they must be reconstructed, and the reconstruction rule is a researcher degree of freedom. We therefore do not defend a single rule. We count under six, and report the spread.

Grouping by transaction, by a one-second gap, or by the whole hour lands within 2% of the same answer (Table 1). Tranching happens *inside* a transaction, not across transactions separated in time, so the choice of rule is not doing the work. We adopt (account, transaction) for the remainder.

One caution earned in the course of this work: a transaction hash is not a liquidation. A single transaction can carry forced closes for several accounts, so grouping by hash alone under-counts. The grouping key must include the account.

Table 1: Count inflation is insensitive to the reconstruction rule. Twelve-hour sample, 24,566 fills.

unit	count	inflation
fills	24,566	1.0×
(account, transaction)	6,412	3.8×
episode, 1 s gap	6,582	3.7×
episode, 5 s gap	6,546	3.8×
episode, 60 s gap	6,533	3.8×
(account, instrument, hour)	6,465	3.8×

5 Count inflation

On the one-year stratified sample, counting fills instead of episodes inflates the event count by a factor of

$$\frac{2{,}010{,}042}{351{,}540} = \mathbf{5.72}.$$

The factor is comparable across segments: 5.78× for majors (289,283 episodes from 1,672,034 fills) and 5.43× for HIP-3 (62,257 from 338,008).

5.1 The inflation is size-dependent

A constant multiplier would be a nuisance parameter. This one is not constant.

Table 2: Tranching grows with episode size. One-year sample, 351,540 episodes.

size bucket	episodes	median fills	mean fills
p0–50 ($0 – $1,117)	175,770	2	2.19
p50–90 ($1,117 – $39,307)	140,616	2	3.94
p90–99 ($39,307 – $549,060)	31,638	8	19.11
p99–100 ($549,060 – $194,115,094)	**3,516**	**72**	**132.27**

The rank correlation between log notional and log fills per episode is +0.545. A typical liquidation is two fills; the largest are a median of 72 and a mean of 132, with a maximum of 4,776.

The consequence is stated most sharply as a share: **the top 1% of episodes generate 23.1% of all fills**. A dataset of fills is therefore not a dataset of liquidations with a scaling factor—it is a dataset in which the largest events have been replicated most.

5.2 Does the 30-second cooldown split episodes?

The partial-liquidation rule means one position's forced close can span several market orders 30 seconds apart, in different transactions. Our (account, transaction) unit would count those separately, so the reported inflation could be too *low*.

We test this on the year sample by measuring the time between consecutive episodes of the same account and instrument (98,983 consecutive pairs):

Chains of two or more episodes within 90 seconds account for 2,200 chains covering 4,879 episodes—**1.4% of the sample**. Merging every such chain into a single episode moves the inflation factor from 5.72× to **5.76**×. The reported figure is therefore mildly conservative, and the choice of unit remains not load-bearing.

Table 3: Time between consecutive episodes of the same account and instrument. The cooldown signature is present and small; the mass is at separations far too long to be one forced close.

separation	pairs	share
$< 5\,\mathrm{s}$	0	0.0%
5–35 s	1,757	1.8%
35–90 s	922	0.9%
90 s – 10 min	2,545	2.6%
$> 10\,\mathrm{min}$	93,759	**94.7%**

The mechanism is nonetheless identifiable in the data, which is a useful check that we are seeing what the documentation describes: chains begin at a median episode size of **$72,326** against **$1,902** for isolated episodes, a factor of 38, consistent with a rule that engages above 100,000 USDC. Only 4.86% of episodes exceed that threshold.

5.3 Concentration

Liquidated notional is concentrated to a degree that compounds the above: the top 1% of episodes carry **67.3%** of notional, the top 5% carry 85.6%, and the top 10% carry 92.6%. The events that dominate the economics are the events the fill-count distorts most.

6 Size compression

Fill-counting also changes the measured size distribution, because an episode of a given size appears as several smaller records.

Table 4: Fill-counting compresses the size distribution. All 2,010,314 fills of the one-year sample; 95% intervals from 20,000 bootstrap draws.

quantile	true episode	observed fill	factor	95% CI
p50	$1,117	$558	2.00×	[1.97, 2.03]
p90	$39,288	$12,046	3.26×	[3.21, 3.31]
p99	$548,922	$119,919	**4.58**×	[4.41, 4.76]
p99.9	$5,834,505	$582,486	**10.02**×	[9.24, 11.21]
maximum	$194,115,094	$10,990,000	17.7×	—

Compression grows monotonically through the distribution, reaching an order of magnitude at the 99.9th percentile (Table 4). The largest episode in the sample, $194,115,094 across 4,776 fills, appears in the fill record as pieces of at most $10.99M.

The cascade-selected sample understated this. An earlier version of this measurement used the twelve-hour sample described in Section 4, two of whose hours had been chosen for containing cascades. It reported 1.6×, 2.1×, 3.4× and 10.9× at the same four quantiles. The year-scale figures are larger at p50, p90 and p99, with the bootstrap interval excluding the old value at each; at p99.9 the two are indistinguishable. This is the same direction the count inflation moved—3.8× on the selected hours against 5.72× on the year—and for the same reason: compression is produced by tranching.

The segments differ, and they cross over. Majors and HIP-3 do not compress alike. Taking the ratio of their factors, majors compress more in the body—1.30 at p90, 95% CI [1.24, 1.36]—and HIP-3 more in the tail—0.82 at p99, CI [0.74, 0.90]—with the crossover between the two. Three of four intervals exclude equality.

This is consistent with, rather than contrary to, the finding that the two segments share a tail *index*. An index is a property of the liquidation mechanism; compression is produced by slicing a position against an order book, and HIP-3 books are thinner. A thin book tranches a large position into more and smaller pieces, which is the direction observed.

7 The episode-size distribution

7.1 What holds

The tail is unambiguously heavy. Against a Pareto reference fitted on the same exceedances, an exponential specification is rejected decisively (Vuong's test [2], $R = +11.96$, $p < 10^{-4}$, in favour of Pareto). But the Pareto fit is itself rejected: a Kolmogorov–Smirnov test with parametric-bootstrap p-values rejects it at every threshold tried, including at the threshold selected by the Clauset–Shalizi–Newman procedure [1] ($p = 0.010$).

Both two-parameter alternatives beat it. Fitting conditional on $x \geq x_{\min}$ and comparing by Vuong's non-nested likelihood-ratio test, lognormal and Weibull each dominate Pareto at **all 14 estimable thresholds**, with $x_{\min}$ ranging from \$14,912 to \$8,845,704 —nearly three orders of magnitude. An exponential never wins at any threshold.

7.2 What does not: the tail cannot be named

Selecting $x_{\min}$ by minimising the KS distance of the power-law fit gives \$560,627 ($k = 3{,}104$, $\hat{\alpha} = 0.960$). A nonparametric bootstrap over 500 resamples puts a 95% interval on that selection of [\$193,877, \$986,963]—**a factor of five**. The threshold is barely identified by the data, which is why we report the ranking across a grid of thresholds rather than at a single selected one.

Across that grid the ranking between the two survivors *reverses*:

Table 5: Which alternative wins depends on where the tail is cut. Vuong R for lognormal against Weibull; $R < 0$ favours Weibull.

$x_{\min}$ range	Vuong R	verdict
\$199k – \$1.33M	-2.74 to -3.67 ($p < 0.01$)	Weibull
\$1.85M – \$8.85M	-1.00 to -1.86 ($p > 0.05$)	indistinguishable
\$14.9k – \$28.3k	$+13.42$ to $+15.92$ ($p \ll 0.001$)	lognormal

Weibull wins in the far tail, lognormal wins decisively deeper in the body, and the crossover falls inside a band where neither is identified. **Any name given to this tail would report the analyst's threshold rather than the distribution.**

We take this to be the substantive finding of the section rather than a gap in it.

7.3 A band where nothing fits

Over $x_{\min} \in$ [\$49k, \$163k], both alternatives converge on a parameter bound—lognormal with $\mu \to -\infty$, Weibull with $\lambda \to 0$. Both limits are the power-law limit of their own family. In that band the extra parameter buys nothing and the models are unidentified, while the power law they are collapsing onto is itself rejected by the KS test.

So over that band *no candidate here describes the data.* A model ranking cannot report this, because a ranking always returns a winner. We flag it because a reader who takes the winning family from a comparison table has no way to see it.

7.4 Withdrawn estimates

Three tail-index estimates were produced in the course of this work—$\alpha \approx 1.15$, then $\alpha \in [0.9, 1.2]$, then a Hill-plot plateau at $\alpha \approx 0.93$—and all three are withdrawn. The Hill estimator [3] estimates a Pareto exponent, and the tail is not Pareto; drift of the Hill index with tail depth, initially read as a measurement defect, is the known signature of a lognormal-like tail. Consequences drawn from those numbers, including "infinite variance" and "the sample mean is not a stable statistic," are withdrawn with them: both are properties of a fitted Pareto with $\alpha < 2$, and that fit is rejected.

The count and compression results in Sections 5–6 are unaffected. They compare two curves computed identically on the same data and assume no parametric form.

8 Why this does not transfer to centralised-exchange feeds

The bias measured here is specific to a *fill*-level record. Centralised venues publish liquidation *orders*, so there is no tranching to inflate. Their feeds carry a different defect.

Table 6: What the major centralised feeds publish, per venue documentation.

venue	stream	unit	limitation
Binance	`forceOrder`	order	only the *largest* order per 1000 ms per symbol
OKX	`liquidation-orders`	order	at most one update per second per contract
Bybit (legacy)	`liquidation`	order	at most 1/s per symbol; deprecated
Bybit (current)	`allLiquidation`	order	complete, pushed every 500 ms

Binance does not publish a sample of liquidations. It publishes a sequence of *per-second maxima*, which is a different statistical object with three consequences:

- **event counts are unrecoverable**—nothing in the feed states how many liquidations a given snapshot suppressed;
- **the body of the distribution is destroyed**—small liquidations never appear at all;
- **the tail index may survive**—block maxima are a valid extreme-value approach, and a block maximum inherits the tail index of its parent distribution.

That last point matters for interpreting existing results: a study of Binance liquidation data can produce a defensible tail estimate and a worthless event count from the same feed. The distinction is between statistics that depend on counting or on the body, which are compromised, and tail-shape statistics, which may not be.

We note this from venue documentation, not from a re-analysis of published work, and make no claim about any specific paper.

9 Implications

For measurement. Any statistic over crypto liquidation data should state its unit. On a fill-level record the correction is a factor of 5.72 on counts, but because the factor is size-dependent it cannot be applied as a scalar: the correct procedure is to reconstruct episodes, which requires an attributed record.

For cascade studies. Cascade detection typically thresholds on event counts or event rates within a window. Both are inflated most during the large episodes that constitute cascades, so a fill-based cascade measure partly detects its own tranching.

For risk models. A size distribution fitted on fills is compressed by 3.4× at the 99th percentile and 10.9× at the 99.9th. A model calibrated on that distribution understates the size of the largest forced closes it is meant to survive.

For the tail literature. We would encourage reporting the model ranking across a range of thresholds rather than at a selected one, and reporting where no candidate fits. In this dataset a single-threshold analysis would have supported naming the tail; the grid shows the name would have been the threshold's.

10 Limitations

- **The p99.9 and maximum rows are thin.** One in a thousand of 351,648 episodes is 352 observations, and the maximum is a single episode against a single fill, quoted without an uncertainty statement.
- **Two collections of the same hours differed by 0.03%** (108 episodes, 272 fills), most likely late-arriving blocks written after the first pass. It changes no figure here, but work needing exact reproducibility should pin an archive snapshot.
- **The sample is stratified, not exhaustive**: four fixed hours per day. Uniform in time, but a cascade falling entirely outside those hours is invisible.
- **One venue, one year.** Whether the inflation factor is stable over time, or trends with venue growth, is untested.
- **The goodness-of-fit bootstrap fixes $x_{\min}$** inside each synthetic sample rather than re-selecting it, which makes the reported p conservative toward rejection.
- **Four candidate families.** A fifth—generalised Pareto, log-gamma—might describe the band where none of ours does. None was tried.
- **Our novelty claim is bounded by a search, not by a proof.** We surveyed the empirical crypto-liquidation literature and found no study that counts liquidation events from a fill-level record, and none that states its counting unit. Absence of a result in a search is weaker than absence of the result. The claim to read is the measurement, not the priority.
- **Two cited works were not read in full.** Lim [9] is behind a registration wall and Zhivkov et al. [10] returned an access error; both are characterised from their abstracts and metadata. Nothing in our argument depends on their contents beyond the fact that they measure in dollars and across venues respectively.

11 Reproducibility

All code, reduced datasets and experimental records are public at `https://github.com/QuasareumGroup/hyperliquid-microstructure`.

Each experiment is pre-registered: hypothesis, predictions and falsification criteria are committed *before* the run, and the commit ordering is checkable in the repository history. Corrections and withdrawals remain in the file that made the original claim rather than being edited away; `experiments/FINDINGS.md` carries the current state of every claim and the reason for each of the fourteen that have been withdrawn or requalified.

Published datasets carry a derived account identifier—a truncated SHA-256 of the address—in place of the address itself. Addresses are public on-chain, so this is not confidentiality; it avoids publishing a compiled map of 151,730 accounts to their losses. Every result reproduces bit for bit on the derived identifiers. The hash is unsalted and deterministic, so a known address can still be tested for membership: these datasets are pseudonymous, not anonymised.

References

[1] A. Clauset, C. R. Shalizi, and M. E. J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009.

[2] Q. H. Vuong. Likelihood ratio tests for model selection and non-nested hypotheses. *Econometrica*, 57(2):307–333, 1989.

[3] B. M. Hill. A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5):1163–1174, 1975.

[4] Z. Cheng, J. Deng, T. Wang, and M. Yu. Liquidation, leverage and optimal margin in bitcoin futures markets. *Applied Economics*, 53(47), 2021. arXiv:2102.04591.

[5] K. Qin, L. Zhou, P. Gamito, P. Jovanovic, and A. Gervais. An empirical study of DeFi liquidations: incentives, risks, and instabilities. In *Proc. 21st ACM Internet Measurement Conference (IMC)*, 2021. arXiv:2106.06389.

[6] C. C. Moallemi and U. Patange. An analysis of fixed-spread liquidation lending in DeFi. SSRN 4803547, 2024.

[7] T. Chitra, N. Thogiti, M. J. P. Trujillo Ramirez, and V. Xu. Autodeleveraging as online learning. arXiv:2602.15182, February 2026.

[8] O. Sepper. Slippage-at-Risk (SaR): a forward-looking liquidity risk framework for perpetual futures exchanges. arXiv:2603.09164, March 2026.

[9] B. C. Lim. Anatomy of a crypto cascade: minute-level evidence from the October 2025 crash. SSRN 6579278.

[10] K. Zhivkov, T. Todorov, and G. Georgiev. Temporal dynamics of market microstructure in cryptocurrency perpetual futures: econometric evidence from centralized and decentralized exchanges. *Journal of Risk and Financial Management*, 14(5):103, 2026.

[11] K33 Research. Cited in: Crypto liquidations may be way worse than data has let on. *Cointelegraph*, August 2024. `https://cointelegraph.com/news/liquidation-data-wildy-underreported-last-3-years-k33-research`.

[12] Hyperliquid. Documentation: liquidations, node data, and the funding mechanism. `https://hyperliquid.gitbook.io/hyperliquid-docs/trading/liquidations`.

[13] Binance. USD-M futures WebSocket streams: liquidation order streams. `https://developers.binance.com/docs/derivatives/usds-margined-futures/websocket-market-streams/Liquidation-Order-Streams`.

[14] OKX. WebSocket API: liquidation orders channel. `https://www.okx.com/docs-v5/en/`.

[15] Bybit. V5 WebSocket public: `allLiquidation` stream. `https://bybit-exchange.github.io/docs/v5/websocket/public/all-liquidation`.

[16] CoinGlass. Liquidation heatmap methodology. `https://docs.coinglass.com/reference/liquidation-aggregate-heatmap`.